**Exp No: 9**

**ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS**

**AIM:**

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

**SOFTWARE REQUIRED:**

- Proteus 8 software
- Arduino IDE

PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
int delay_time=200;

void setup() {
//configure pin2 as an input and enable the internal pull-up resistor
pinMode(12, INPUT_PULLUP);
pinMode(1,OUTPUT);

pinMode(2,OUTPUT);
pinMode(3,OUTPUT);
pinMode(4,OUTPUT);
pinMode(5,OUTPUT);
pinMode(6,OUTPUT);
pinMode(7,OUTPUT);

}
void loop() {
//read the pushbutton value into a variable
int sensorVal = digitalRead(12);
```

```
if (sensorVal == LOW) {
x0=true;
}
while(x0){
zero();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x1=true;
x0=false;
}
}
while(x1){
one();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x2=true;
x1=false;
}

}
while(x2){
two();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x3=true;
x2=false;
}
}
while(x3){
three();
```

```
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x4=true;
x3=false;
}
}
while(x4){
four();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x5=true;
x4=false;
}
}
while(x5){
five();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x6=true;

x5=false;
}
}
while(x6){
six();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x7=true;
x6=false;
}
```

```
}
while(x7){
seven();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x8=true;
x7=false;
}
}
while(x8){
eight();
sensorVal = digitalRead(12);
if (sensorVal == LOW) {
x9=true;
x8=false;
}
}
while(x9){
nine();
sensorVal = digitalRead(12);

if (sensorVal == LOW) {
x0=true;
x9=false;
}
}

}

void zero()
```

```
{
for(int i=1;i<7;i++)
{
digitalWrite(i,HIGH);
digitalWrite(7,LOW);
}
delay(delay_time);
}

void one()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,LOW);
delay(delay_time);
}

void two()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,LOW);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
```

```
delay(delay_time);
}

void three()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
digitalWrite(7,HIGH);
delay(delay_time);
}

void four()
{
digitalWrite(1,LOW);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);

digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void five()
{
```

```
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void six()
{
digitalWrite(1,HIGH);
digitalWrite(2,LOW);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void seven()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,LOW);
digitalWrite(5,LOW);
digitalWrite(6,LOW);
```

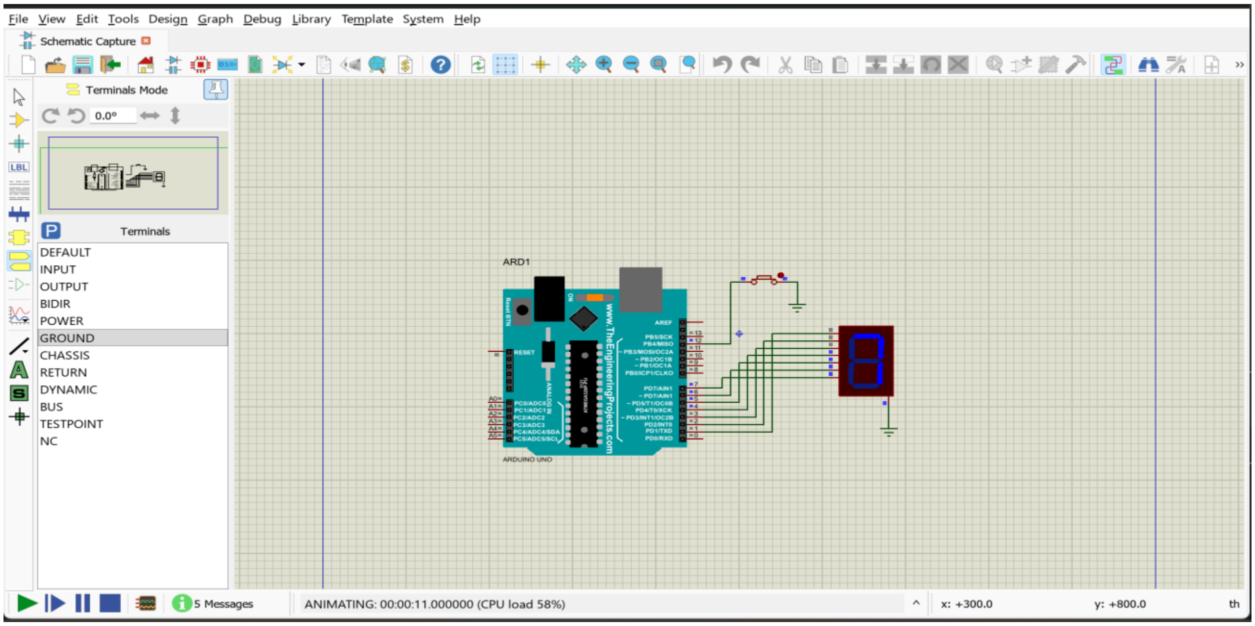
```
digitalWrite(7,LOW);
delay(delay_time);
}

void eight()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);
digitalWrite(5,HIGH);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}

void nine()
{
digitalWrite(1,HIGH);
digitalWrite(2,HIGH);
digitalWrite(3,HIGH);
digitalWrite(4,HIGH);

digitalWrite(5,LOW);
digitalWrite(6,HIGH);
digitalWrite(7,HIGH);
delay(delay_time);
}
```

**Output:**


File View Edit Tools Design Graph Debug Library Template System Help
Schematic Capture
Terminals Mode
0.0°
Terminals
DEFAULT
INPUT
OUTPUT
BIDIR
POWER
GROUND
CHASSIS
RETURN
DYNAMIC
BUS
TESTPOINT
NC
ARD1
www.TheEngineeringProjects.com
ARDUINO UNO
5 Messages
ANIMATING: 00:00:11.000000 (CPU load 58%)
x: +300.0
y: +800.0